

3TC-Machine Learning For Prediction

Group members:

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Model 1: Logistic Regression

Evaluation metrics: LR-Accuracy

Classification Report:

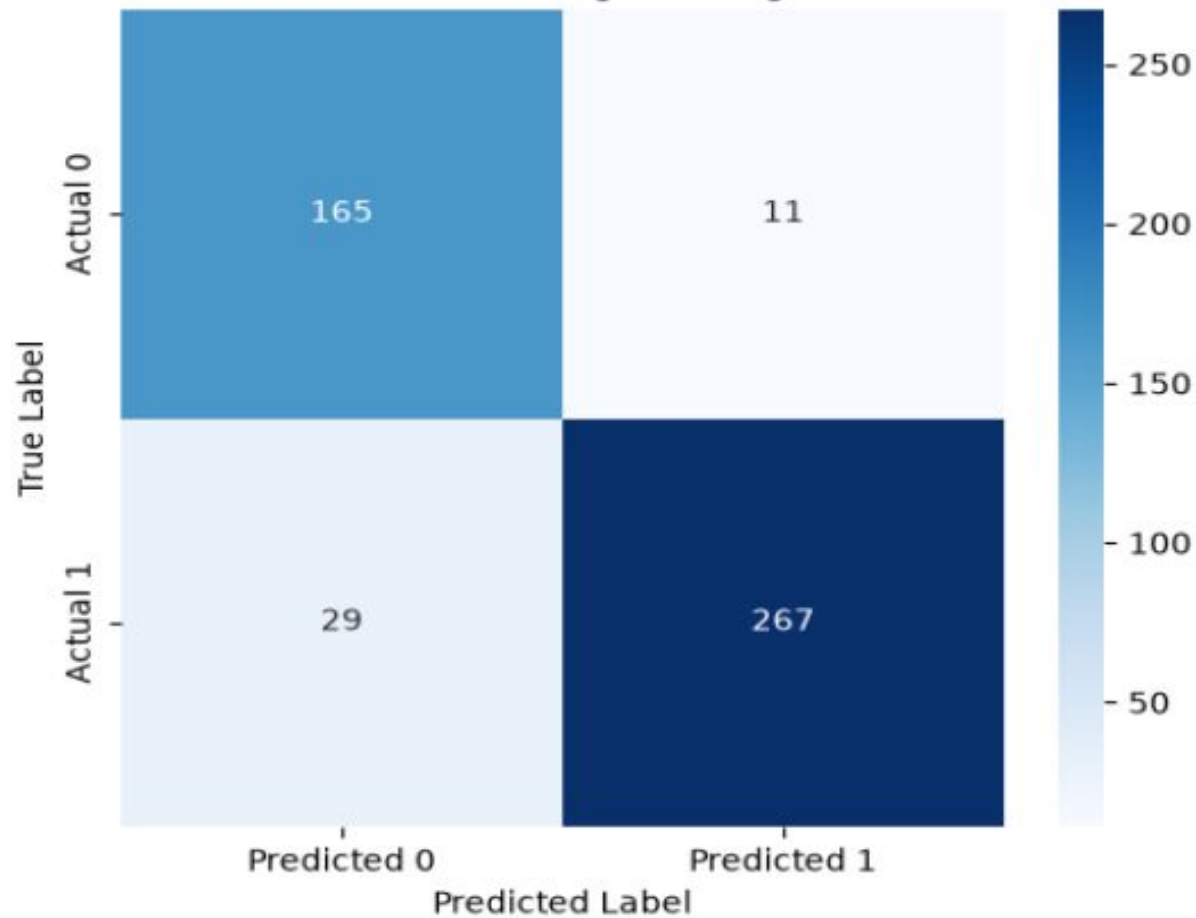
	precision	recall	f1-score	support
0	0.85	0.94	0.89	176
1	0.96	0.90	0.93	296
accuracy			0.92	472
macro avg	0.91	0.92	0.91	472
weighted avg	0.92	0.92	0.92	472

Evaluation metrics : PRECISION & RECALL

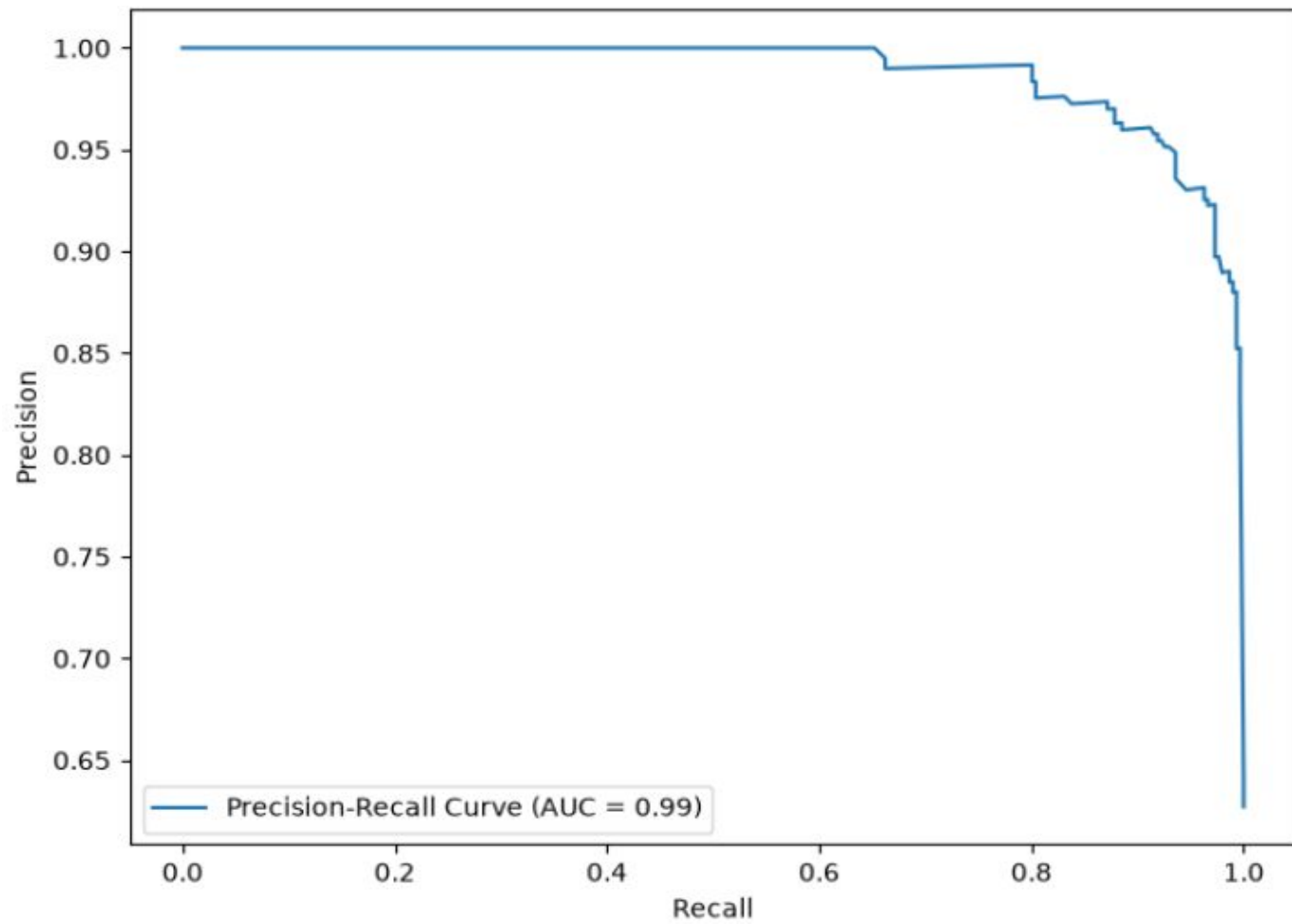
Classification Report:

	precision	recall	f1-score	support
0	0.85	0.94	0.89	176
1	0.96	0.90	0.93	296
accuracy			0.92	472
macro avg	0.91	0.92	0.91	472
weighted avg	0.92	0.92	0.92	472

Confusion Matrix - Logistic Regression

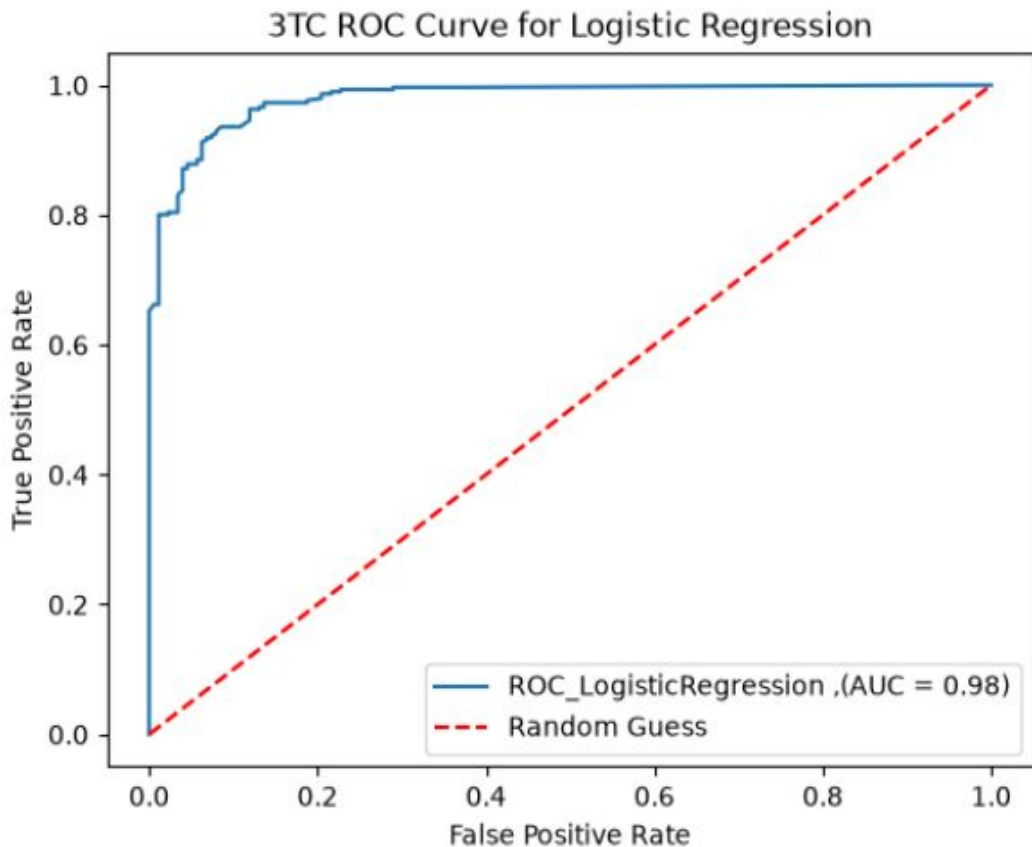


Precision- Recall & ROC AUC curves



Precision, Recall, F1 score, precision- Recall & **ROC AUC curves**

- Data set is balanced - (our dataset is imbalanced)
- False positives and False negatives are of equal importance #Critical for our model



Model 2: Random Forest Classifier

Evaluation metrics: RF-Accuracy

--- Classification Report ---

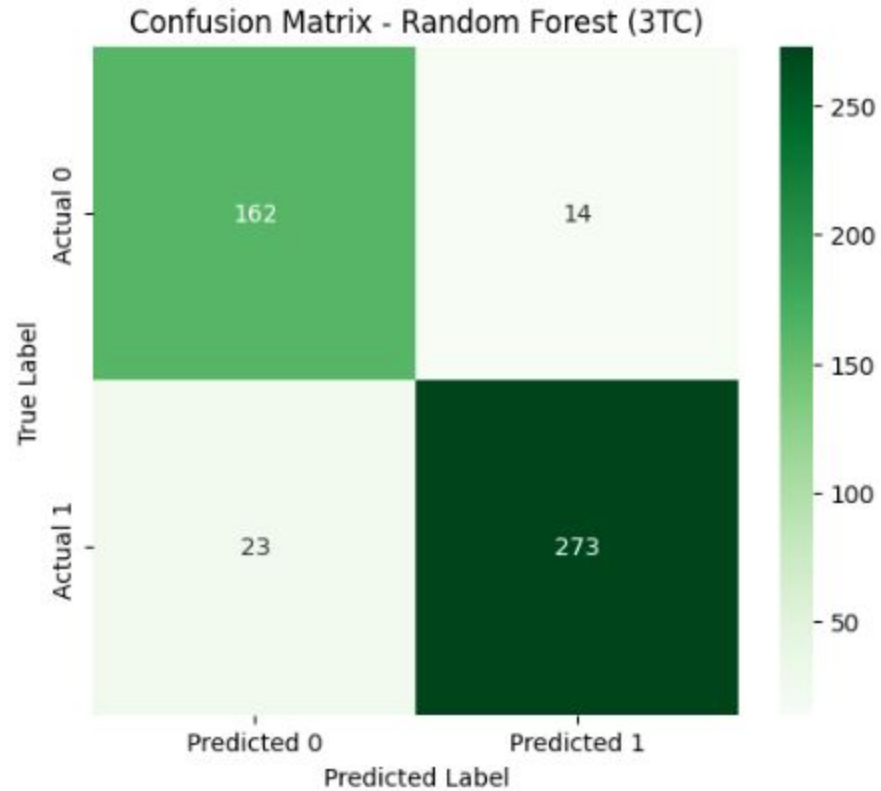
	precision	recall	f1-score	support
0	0.88	0.92	0.90	176
1	0.95	0.92	0.94	296
accuracy			0.92	472
macro avg	0.91	0.92	0.92	472
weighted avg	0.92	0.92	0.92	472

Evaluation metrics :RF PRECISION & RECALL

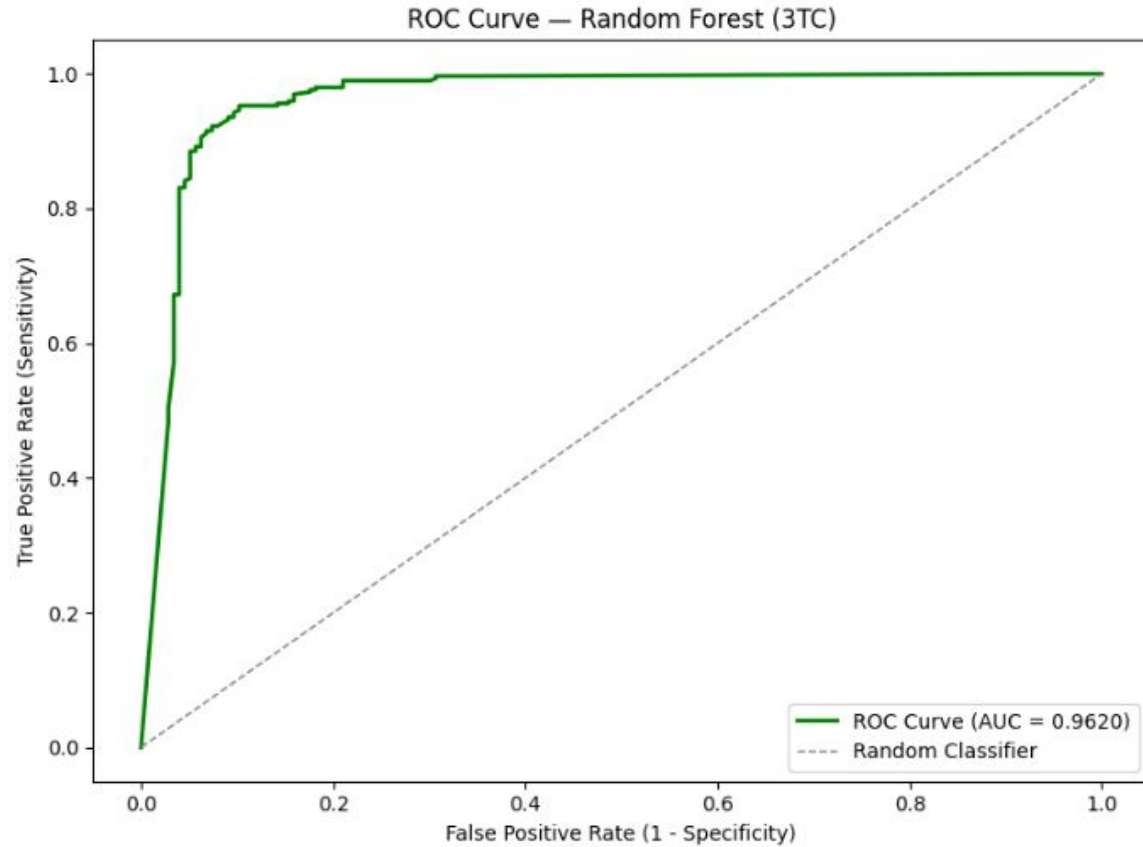
--- Classification Report ---

	precision	recall	f1-score	support
0	0.88	0.92	0.90	176
1	0.95	0.92	0.94	296
accuracy			0.92	472
macro avg	0.91	0.92	0.92	472
weighted avg	0.92	0.92	0.92	472

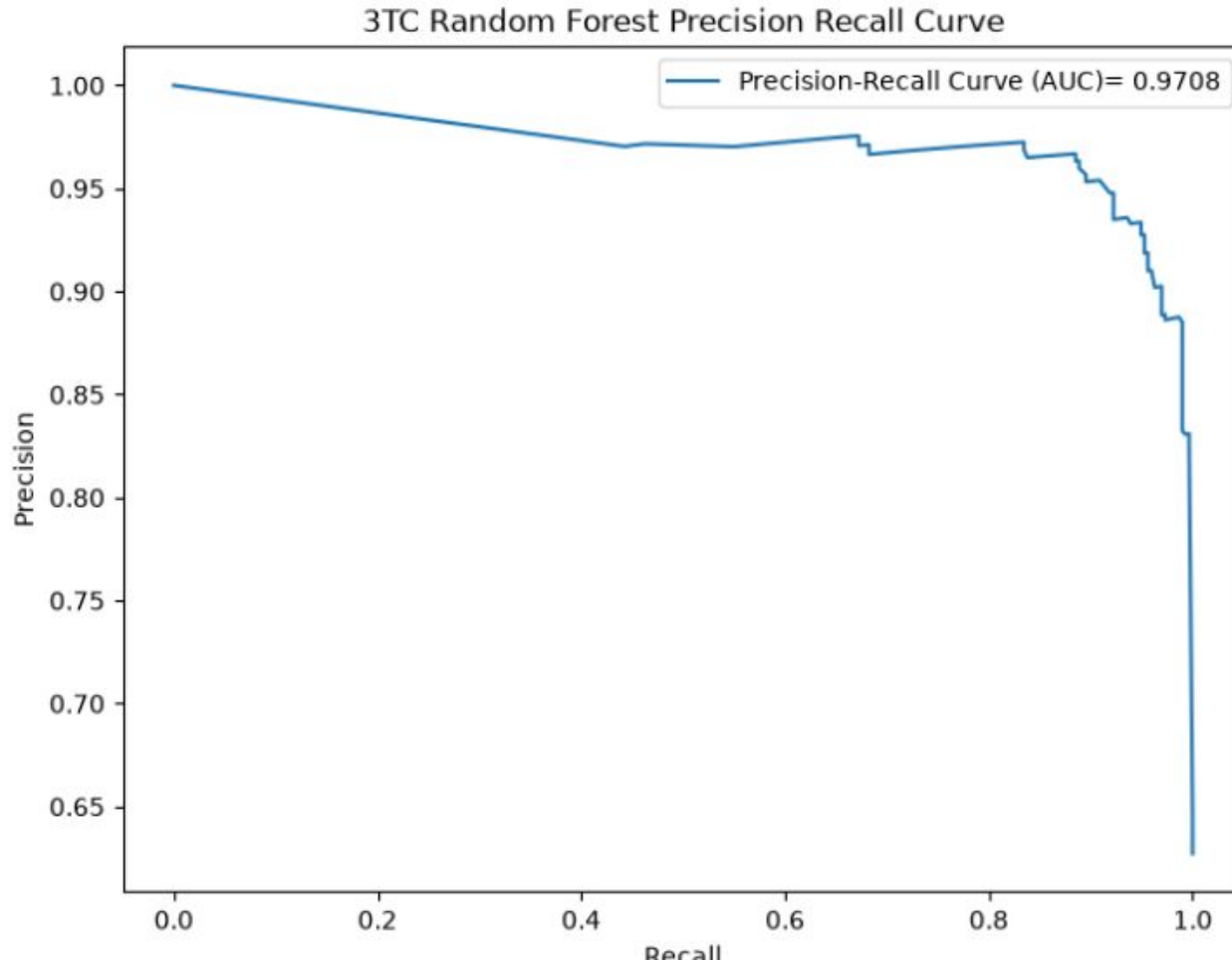
Confusion Matrix - Random Forest



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves



Model 3: Support Vector Machine

Evaluation metrics: SVM-Accuracy

SVM Test Classification Report ---

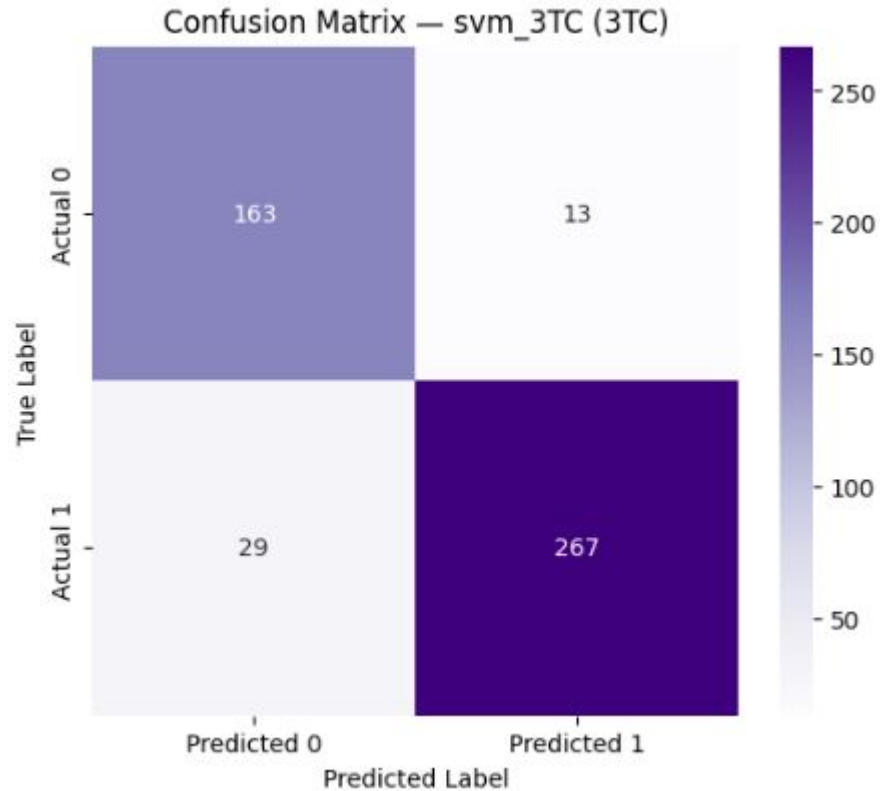
	precision	recall	f1-score	support
0	0.85	0.93	0.89	176
1	0.95	0.90	0.93	296
accuracy			0.91	472
macro avg	0.90	0.91	0.91	472
weighted avg	0.91	0.91	0.91	472

Evaluation metrics: SVM-PRECISION & RECALL

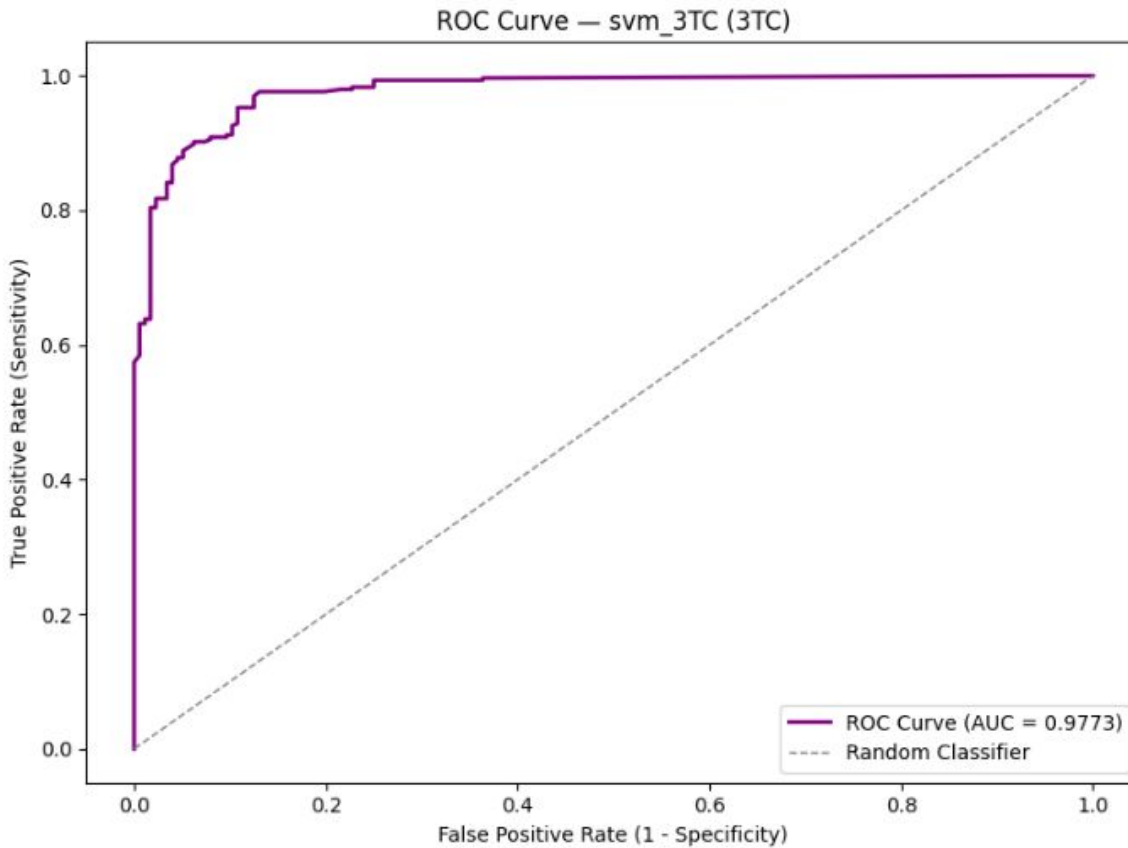
SVM Test Classification Report ---

	precision	recall	f1-score	support
0	0.85	0.93	0.89	176
1	0.95	0.90	0.93	296
accuracy			0.91	472
macro avg	0.90	0.91	0.91	472
weighted avg	0.91	0.91	0.91	472

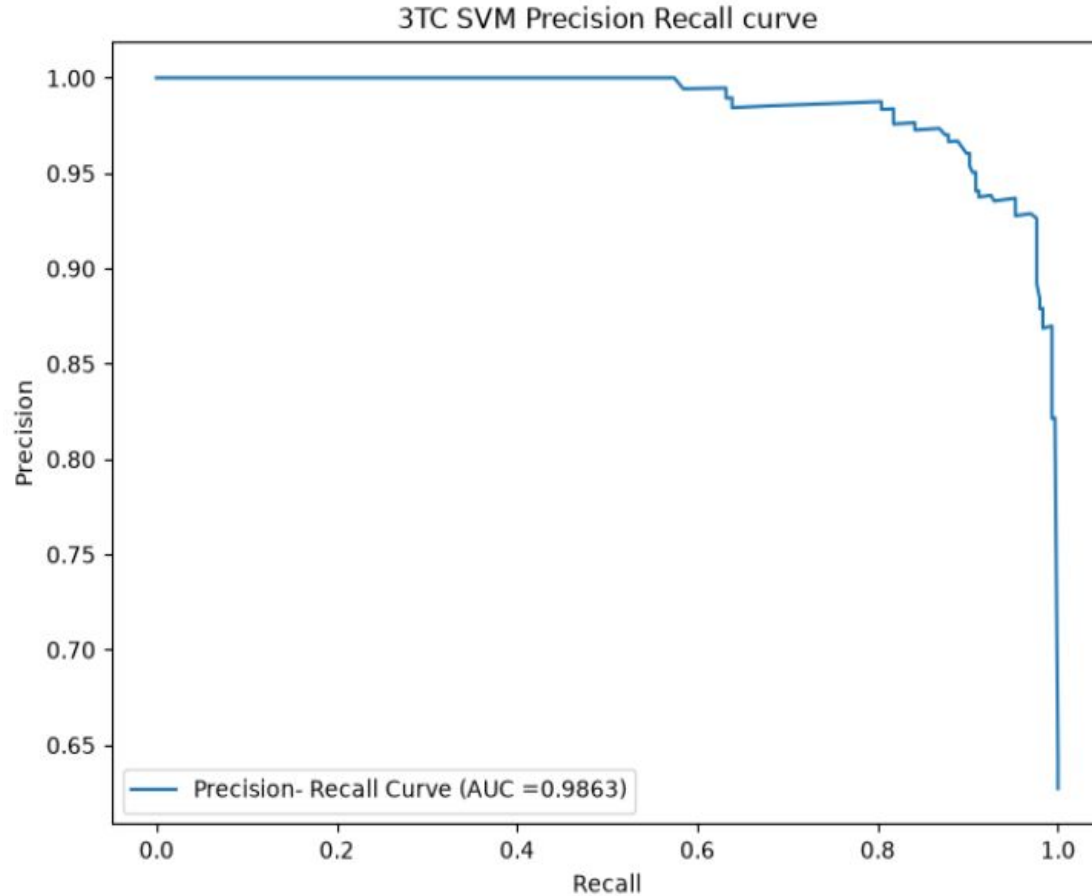
Confusion Matrix - Support Vector Machine



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves



Model 4: Extreme Gradient Boost

Evaluation metrics: **XGB-Accuracy**

--- XgBoost Test Classification Report ---

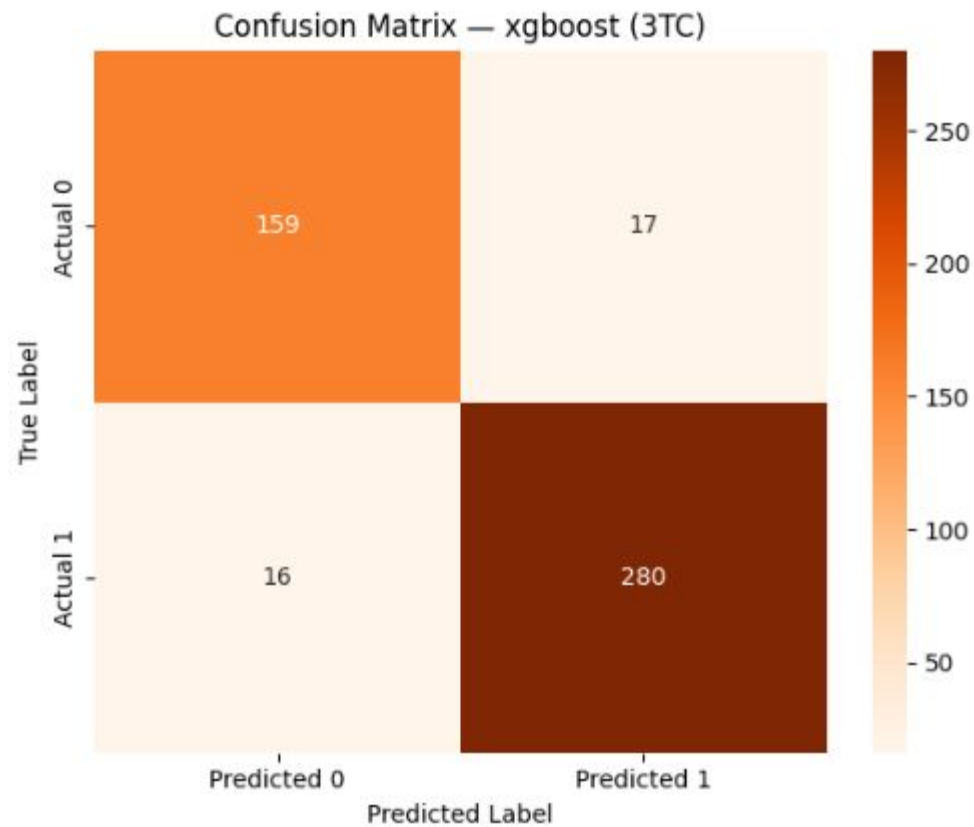
	precision	recall	f1-score	support
0	0.91	0.90	0.91	176
1	0.94	0.95	0.94	296
accuracy			0.93	472
macro avg	0.93	0.92	0.93	472
weighted avg	0.93	0.93	0.93	472

Evaluation metrics: XGB-Accuracy

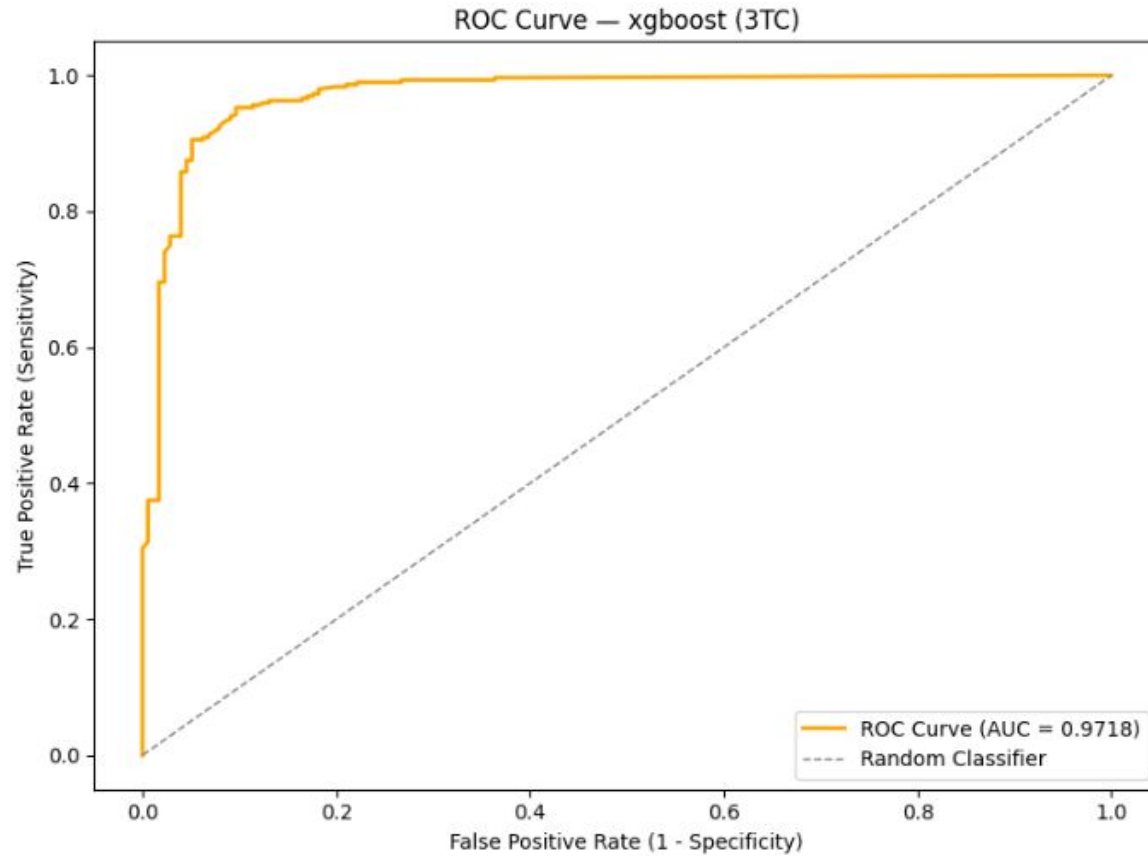
--- XgBoost Test Classification Report ---

	precision	recall	f1-score	support
0	0.91	0.90	0.91	176
1	0.94	0.95	0.94	296
accuracy			0.93	472
macro avg	0.93	0.92	0.93	472
weighted avg	0.93	0.93	0.93	472

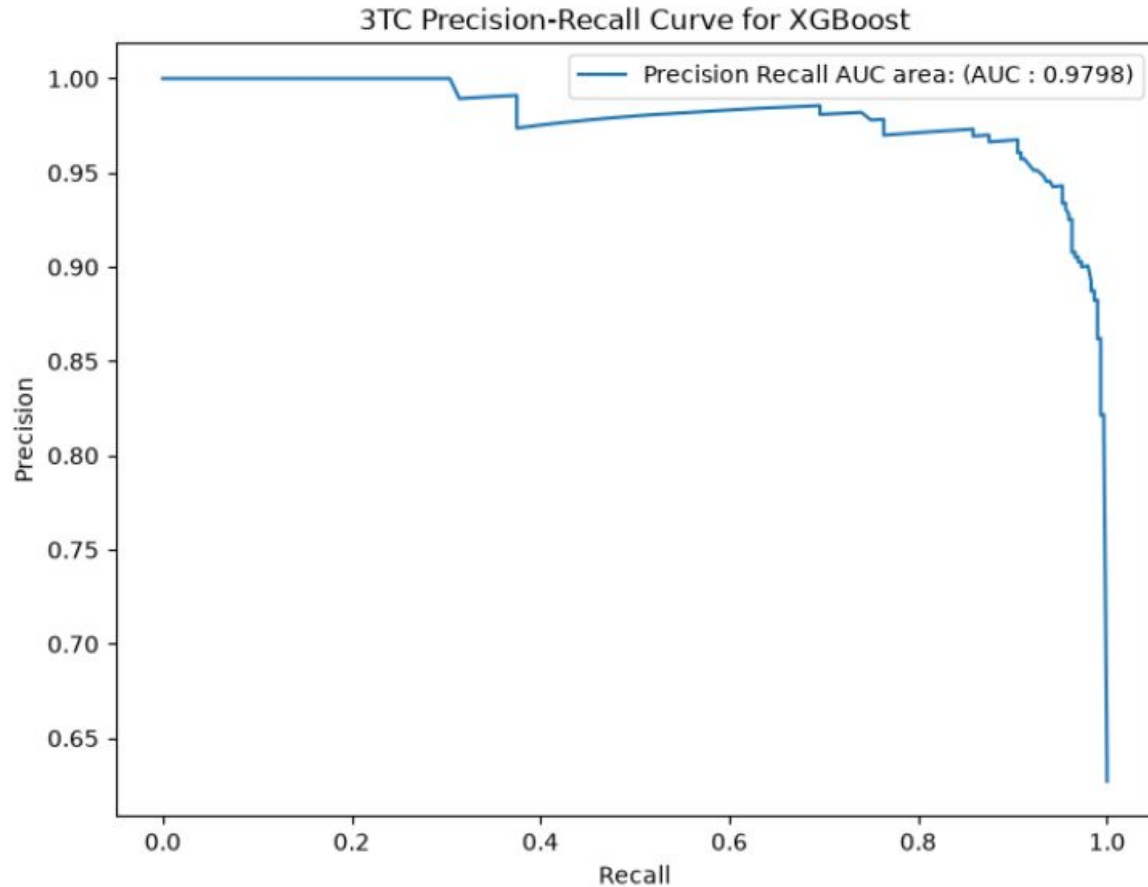
Confusion Matrix - XGBOOST



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves



Comparison (Macro -avg) - 3TC

Model	Precision	Recall	F1 score
LR	0.91	0.92	0.91
RF	0.91	0.92	0.92
SVM	0.90	0.91	0.91
XGB	0.93	0.92	0.93

Model Ranking by MCC (3TC)

Rank	Model	MCC
1	XGB	0.8503
2	RF	0.8348
3	LR	0.8251
4	SVM	0.8152

Best model : XGB

ROC Curves 3TC

